HOMEWORK #1

F-16 Plant Model and Aerodynamic Coefficients

UUM 523E – Advanced Flight Dynamics · Fall 2026-2027

Homework at a glance	
Due date	Oct 30, 2026
Points	Problem 1: 80 points; Problem 2: 20 points (part a: 10, part b: 10)
Report	At most 8 pages, submitted through Ninova
Simulink models	Saved in MATLAB R2023a
Results file	HW1_Group<No>.mat, one per group (see Problem 1)
Submission	One submission per group: report, code, Simulink models and results file

Late submissions

A penalty of 10 points per day is applied to late submissions, and late submissions are accepted for a maximum of 3 days after the due date.

Provided material

In the aerodynamic data provided, you will find the aerodynamic data (f16_AerodynamicData.data) and the corresponding breakpoints (f16_AerodynamicData.BP) given separately for each aerodynamic coefficient. The engine data is also provided in "f16_EngineData.mat". Make sure that you use the correct breakpoints when you create your model.

Please refer to the reference paper "*NASA Technical Paper 1538 - Simulator Study of Stall/Post-Stall Characteristics of a Fighter Airplane With Relaxed Longitudinal Static Stability*" for the details of the aerodynamic modeling.

Modeling conventions

- **Atmosphere:** use the standard atmosphere model of the F-16 model in Stevens, Lewis, and Johnson [2], Appendix A.6 (p. 715). That model is written in English units; implement it in SI units (altitude in *m*, temperature in *K*, density in *kg/m³*, airspeed in *m/s*, pressure in *Pa*) by converting its constants.
- **Axes:** give the aerodynamic forces (X, Y, Z) and moments (L, M, N) in body axes, with the moments taken about the center of gravity at $x_{cg} = 0.35\bar{c}$.
- **Leading-edge flap:** in the validation cases, use the δ_{lef} values given in Table 1 directly; do not use the leading-edge flap schedule.
- **Unit conversions:** the aerodynamic and engine data are in English units. Convert to SI units with 1 *ft* = 0.3048 *m* and 1 *lbf* = 4.448222 *N*.

1. Problems

Problem 1 [80 points] – F-16 plant model in Simulink

Construct a plant model of the F-16 aircraft in Simulink. The plant model should include the following subsystem models:

- Aerodynamics
- Propulsion
- Mass & Inertia
- Atmosphere
- Actuators
- Equations of Motion

Aerospace Blockset: for validation only

Please model each subsystem without using "Aerospace Blockset". **Simulink models that contain Aerospace Blockset blocks will not be accepted.** Aerospace Blockset may be used only to validate the outputs of your own subsystem models. The evaluation will be based on your knowledge of how to construct the subsystem models.

Problem 1 – Validation of the aerodynamics subsystem

Validate your aerodynamics subsystem model for the ten cases in Table 1. For each case, calculate the aerodynamic coefficients $(C_X, C_Y, C_Z, C_l, C_m, C_n)$, the aerodynamic forces (X, Y, Z) in N and the aerodynamic moments (L, M, N) in $N{\cdot}m$.

Flight conditions: 0.4 Mach, 10000 *ft*; $x_{cg} = 0.35\overline{c}$

Table 1. Aerodynamic validation cases

CASE	Alpha (*deg*)	Beta (*deg*)	p (*deg/s*)	q (*deg/s*)	r (*deg/s*)	δ_a (*deg*)	δ_h (*deg*)	δ_r (*deg*)	δ_{lef} (*deg*)	δ_{sb} (*deg*)
1	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	60
3	20	0	0	0	0	0	0	0	0	0
4	0	12.5	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	-15	0	0	0
6	0	0	0	0	0	0	0	0	25	0
7	0	0	0	0	0	10	0	0	0	0
8	0	0	0	0	0	0	0	-15	0	0
9	5.16	0	0	0	0	0	-0.47	0	7.55	0
10	5.95	0	-0.53	1.21	2.09	-0.055	-0.62	-0.33	8.65	0

Required output. Store the results of all ten cases in `AeroResults` (see "Results file" below).

Problem 1 – Validation of the propulsion subsystem

Validate your propulsion subsystem model for the five cases in Table 2. For each case, calculate the steady-state thrust in *N*.

Table 2. Propulsion validation cases

CASE	Mach	Altitude (*ft*)	$\delta_{Throttle}$ (%)
1	0.25	5000	10
2	0.65	25000	77
3	0.9	45000	95
4	0.4	10000	16.66
5	0.4	10000	19.77

Required output. Store the thrust of all five cases in `ThrustResults` (see "Results file" below).

Results file (one per group)

Each group submits one MAT-file named `HW1_Group<No>.mat`, where <No> is the group number. The file must contain exactly the two variables below, saved as double arrays:

Variable	Size	Content
`AeroResults`	10 × 12	Row k: case k of Table 1. Columns, in this order: $C_X, C_Y, C_Z, C_l, C_m, C_n, X, Y, Z, L, M, N$
`ThrustResults`	5 × 1	Row k: case k of Table 2. Steady-state thrust in *N*.

Coefficients are dimensionless, forces are in *N* and moments are in *N·m*. Do not change the order of the rows or columns, and do not add other variables, so that the results of all groups can be compared directly. For example:

```
save('HW1_Group<No>.mat', 'AeroResults', 'ThrustResults')
```

Problem 2, part a [10 points] – Lift and drag coefficients

Calculate the lift coefficient (C_L) and the drag coefficient (C_D) for the following conditions:

i. When $\beta = 0°, \delta_h = 0°, \delta_{lef} = 0°, \delta_{sb} = 0°$.
ii. When $\beta = 0°, \delta_h = 0°, \delta_{lef} = 0°, \delta_{sb} = 60°$.
iii. When $\beta = 0°, \delta_h = 0°, \delta_{lef} = 25°, \delta_{sb} = 0°$.
iv. When $\beta = 15°, \delta_h = 0°, \delta_{lef} = 0°, \delta_{sb} = 0°$.
v. When $\beta = 0°, \delta_h = -25°, \delta_{lef} = 0°, \delta_{sb} = 0°$.

Required output. Plot the results with respect to the angle of attack. Comment on the contribution of the parameters to the lift and drag coefficients for the given conditions.

Problem 2, part b [10 points] – Pitching moment coefficient

The aerodynamic pitching moment coefficient of the F-16 is calculated by

$$\begin{aligned} C_{m,t} &= C_m(\alpha,\beta,\delta_h)\eta_{\delta_h}(\delta_h) + \frac{q\bar{c}}{2V_T}\left[C_{m_q}(\alpha) + \Delta C_{m_q,lef}(\alpha)\left(1-\frac{\delta_{lef}}{25}\right)\right] \\ &+ C_{Z,t}\left(x_{cg,ref} - x_{cg}\right) + \Delta C_{m,lef}\left(1-\frac{\delta_{lef}}{25}\right) + \Delta C_{m,sb}(\alpha)\left(\frac{\delta_{sb}}{60}\right) \\ &+ \Delta C_m(\alpha) + \Delta C_{m,ds}(\alpha,\delta_h) \end{aligned}$$

where "$C_{m,t}$" is the total pitching moment coefficient and

$$\Delta C_{m,lef} = C_{m,lef}(\alpha,\beta) - C_m(\alpha,\beta,\delta_h = 0°)$$

i. Plot the variation of $C_{m,t}$ with α for $\beta = [-30,-15,-10,-6,0,6,10,15,30]$ at the reference c.g. location ($0.35\bar{c}$) when $\delta_h = 0°, \delta_{lef} = 0°, \delta_{sb} = 0°$.

Required output. Comment on the longitudinal static stability for the given sideslip angle range. Then, calculate C_{m,t_α} over the range of $\alpha = [-20°:60°]$. Plot C_{m,t_α} with respect to the angle of attack.

2. Reference material

[1] L. T. Nguyen, M. E. Ogburn, W. P. Gilbert, K. S. Kibler, P. W. Brown, and P. L. Deal, *Simulator Study of Stall/Post-Stall Characteristics of a Fighter Airplane With Relaxed Longitudinal Static Stability*, NASA Technical Paper 1538, 1979.

Use for this homework: details of the aerodynamic modeling.

[2] B. L. Stevens, F. L. Lewis, and E. N. Johnson, *Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems*, 3rd ed., John Wiley & Sons, Hoboken, NJ, 2016.

Use for this homework: standard atmosphere model, Appendix A.6, to be implemented in SI units (see "Modeling conventions").

3. Deliverables

This homework is group work. Each group makes one submission containing all of the following:

- **Report:** Please prepare a report of at most 8 pages for the homework and submit it through Ninova.
- **Code:** All the code that you have prepared should also be submitted. The submitted code should work and be easy to follow.
- **Readme:** Please prepare a "readme" file for the code if necessary.
- **Simulink models:** Please submit your Simulink models saved in MATLAB R2023a.
- **Results file:** One MAT-file per group, `HW1_Group<No>.mat`, in the layout given in Problem 1.